

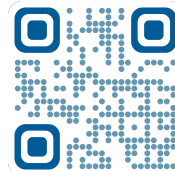
JORDAN DOYLE

Research Software Engineer in Automated Software Testing

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Kildare, Ireland



INTRODUCTION

I recently defended my Ph.D. thesis with SFI Lero and the School of Computer Science in University College Dublin. I am currently a Research Software Engineer in the School of Computer Science and Statistics at Trinity College Dublin, where I am also a member of the Complex Software Laboratory, SFI Lero, and I work in collaboration with Fidelity Investments Ireland. I am looking for a new challenge in the area of Software Engineering as well as new and exciting opportunities to expand my knowledge and expertise.

EDUCATION

Ph.D. in Computer Science

University College Dublin

September 2017 – January 2024

Dublin, Ireland

I designed, implemented, and evaluated an automated model-based approach to generate test inputs for Android mobile applications from a comprehensive model created using both static and dynamic analysis. I also published and presented three research papers at highly regarded conferences.

Java

Python

C

Bash

Android

Appium

Docker

LaTeX



Thesis

Automated Model-based Interface Test Generation for Mobile Applications

B.Sc. in Computer Science (2:1)

University College Dublin

September 2013 – August 2017

Dublin, Ireland

Designed and implemented an end-to-end eReceipt delivery system consisting of a backend database and server, application layer protocol, and a mobile frontend application.

Java

Python

Ruby

C

iOS

Android

SQL

Git



Final Year Project

Green Receipt Manager: An eReceipt management and delivery system

PUBLICATIONS



PADRAIG: Precise Android Automated Input Generation. Doyle, J., Laurent, T., Ventresque, A. 24th IEEE International Conference on Software Quality, Reliability, and Security (QRS) 2024



Modelling Android applications through static analysis and systematic exploratory testing. Doyle, J., Laurent, T., Ventresque, A. 10th IEEE International Conference on Dependable Systems and Their Applications (DSA) 2023



Improving Mobile User Interface Testing with Model Driven Monkey Search. Doyle, J., Saber, T., Arcaini, P., Ventresque, A. 15th IEEE Conference on Software Testing, Verification and Validation Workshops (ICSTW) 2021

EXPERIENCE

Research Software Engineer

Complex Software Laboratory @ Trinity College Dublin

April 2024 – Ongoing

Dublin, Ireland

I am collaborating with Fidelity Investments Ireland on automated testing solutions to detect accessibility defects in web applications, and reduce the manual testing required by test engineers. I am evaluating the usability of these solutions using the company's existing CI/CD infrastructure and comparing them with existing testing methods.

Java

Selenium WebDriver

Selenium Grid

Cucumber

Tutor, TA, & Demonstrator

School of Computer Science @ University College Dublin

September 2017 – January 2024

Dublin, Ireland

I assisted the teaching of undergraduate and master's level students in Functional Programming, Operating Systems, and Software Engineering. As a Teaching Assistant (TA), I coordinated teams of up to 10 demonstrators for modules of

approximately 200 students. I organised and supervised the grading of student assignments, and module practicals. I was recognised as a distinguished TA for the 2023-24 academic year. As a tutor in the Computer Science Support Centre, I provided one-on-one assistance to students struggling with coursework from any module in Computer Science.

Software Engineering (Java)

Operating Systems (Bash)

Functional Programming (Scheme)

Test Automation Intern

Movidius @ Intel Ireland

January 2019 – June 2019

Kildare, Ireland

I researched integration testing methods for newly developed Vision Processing Units (VPU's) and their accompanying development API platform. This required the development and configuration of a custom CI/CD pipeline that included both functional and performance tests with direct access to in-development VPU's and dependency hardware.

C

CMake

GitLab

CICD

Git

Software Engineering Intern

Insight @ University College Dublin

June 2017 - August 2017

Dublin, Ireland

I extended the features and improved the performance of an Android application. The application was designed to guide recent knee or hip replacement patients through recovery exercises with real time visualisation and feedback using a Unity avatar. The collection of physiotherapy research data for Ph.D. students was also a key part of the project.

Java

Android

C#

Unity

Git

Software Engineering Intern

Bruwind @ Vrije Universiteit Brussels

June 2016 - August 2016

Brussels, Belgium

I extended the functionality and features of a web-based application that monitored, analysed and reported the stability of turbine farms in real time, while also collecting data for targeted design optimisations and life-time estimation of turbine substructures.

HTML

CSS

JavaScript

Java

JSF

PrimeFaces

Apache Cassandra

Python

Software Engineering Intern

Manufacturing @ Intel Ireland

June 2014/15 - August 2014/15

Kildare, Ireland

I built a prototype sensor platform using an Intel Galileo Gen 2 board and Arduino sensor modules for deployment in Intel's silicon manufacturing environment. The platform collected environmental data for storage in an SQL database and a web application provided real time monitoring and visualisation of the data for the detection of manufacturing faults. I also developed web applications to display, filter, and update data within relational databases.

HTML

CSS

JavaScript

Python

ASP

VBScript

SQL

C

Arduino

SVN

Git

SKILLS

Software Engineering

Object-Orientated Programming

Functional Programming

Procedural Programming

Web Development

Test Automation

Static Analysis

Dynamic Analysis

Team Coordination & Organisation

Leadership

Research & Development

Technical & Scientific Writing

Presentation & Communication